

How to Make AI Understand Your Data

Beyond simple RAG: context engineering, ontologies, GraphRAG and routing for accuracy, cost and scale

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Why AI agents fail in oil & gas operations

1

Wrong answers

Stale indexes and missing live-data queries omit current work orders, MOC approvals and well status; the model may guess

2

No trust or visibility

Users can't see what data an agent used, what it cost, or why it trusted it

3

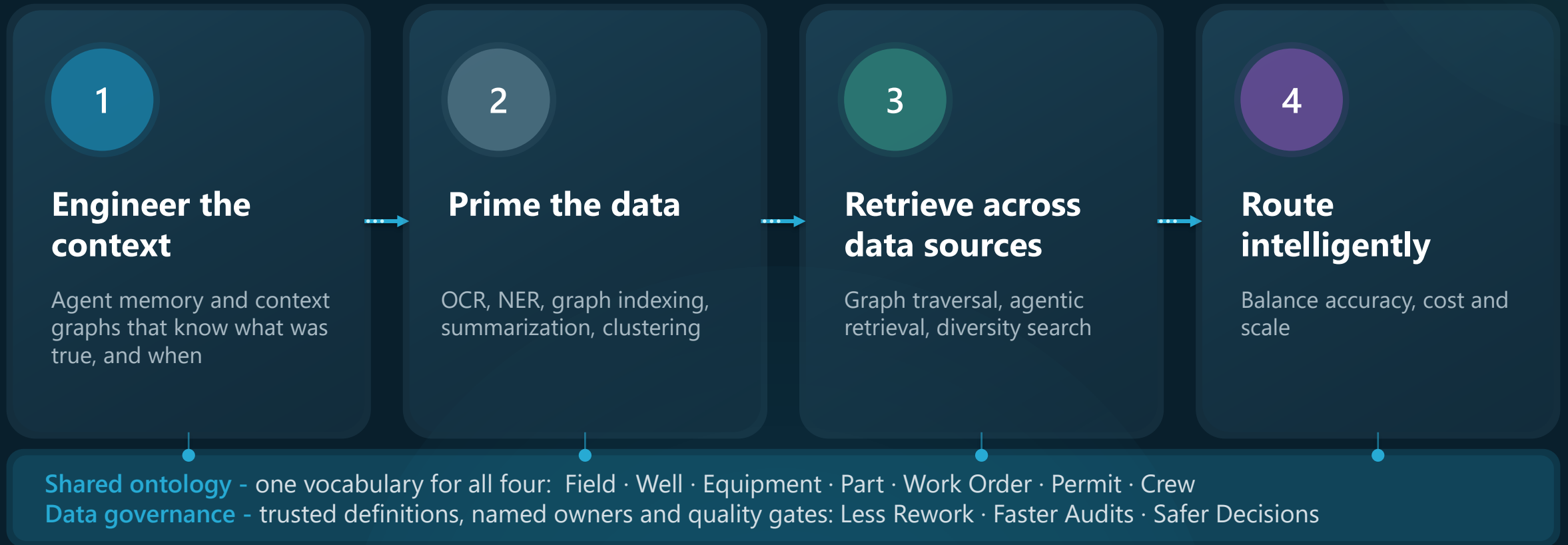
Repeated analysis

Without reusable, validated insights, each run repeats analysis (such as compressor failure): more waiting, more cost

The O&G reality

Truth is scattered across SCADA, historians, ERP, EAM, GIS, well files, drawings and SharePoint. It changes constantly and sits behind strict access controls, so your AI simply doesn't understand your data

Four disciplines that make AI understand your data



Accuracy comes from better context, not just better models.

From retrieval to context engineering

Basic RAG and Prompt Engineering

One search index

Stateless prompts

Chunks without relationships

Only the "latest" version

Stuff the context window

Context and Harness Engineering

Many sources, chosen per question

Short- and long-term agent memory

Ontology-typed entities and relationships

What was true, and when we knew it

Budgeted context: select, compress, cite

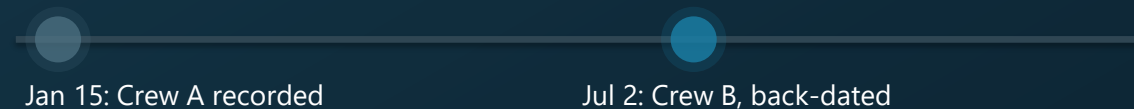
Agent memory on a bi-temporal context graph

Who owned maintenance on compressor C-101?

WHEN IT WAS TRUE — in the field



WHEN WE LEARNED IT — in the system



As believed on Jun 30

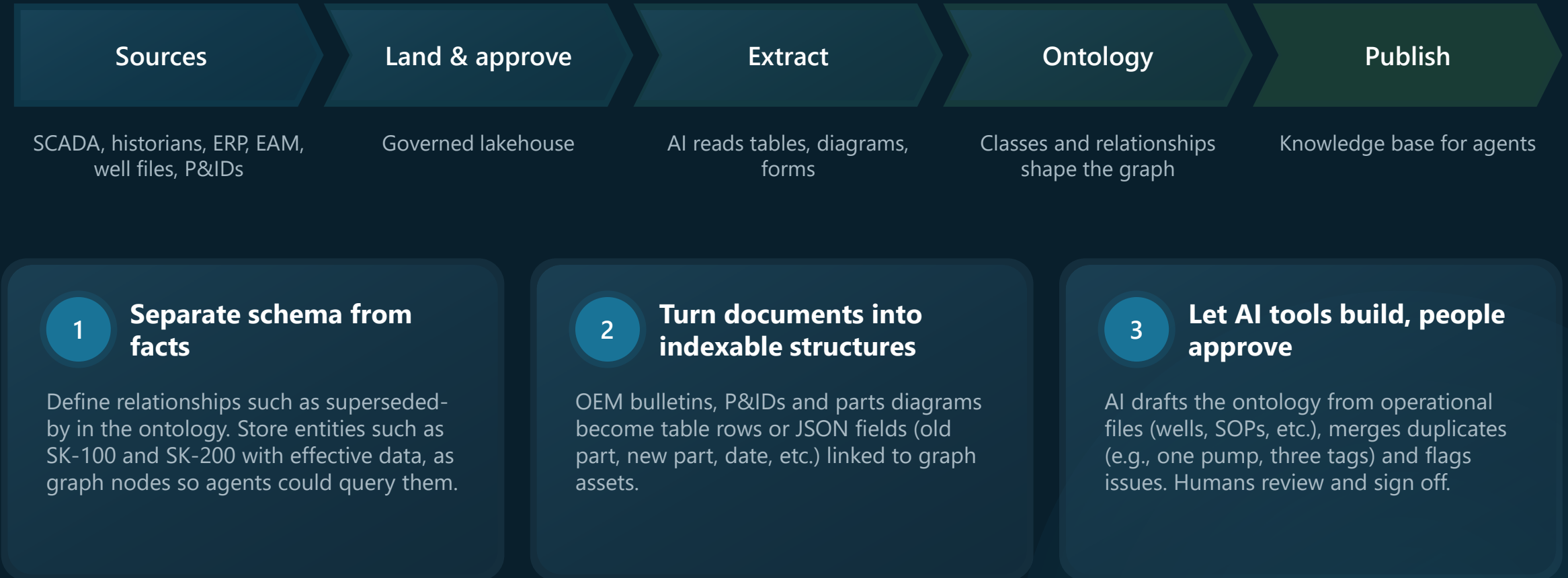
Crew A

As known today

Crew B

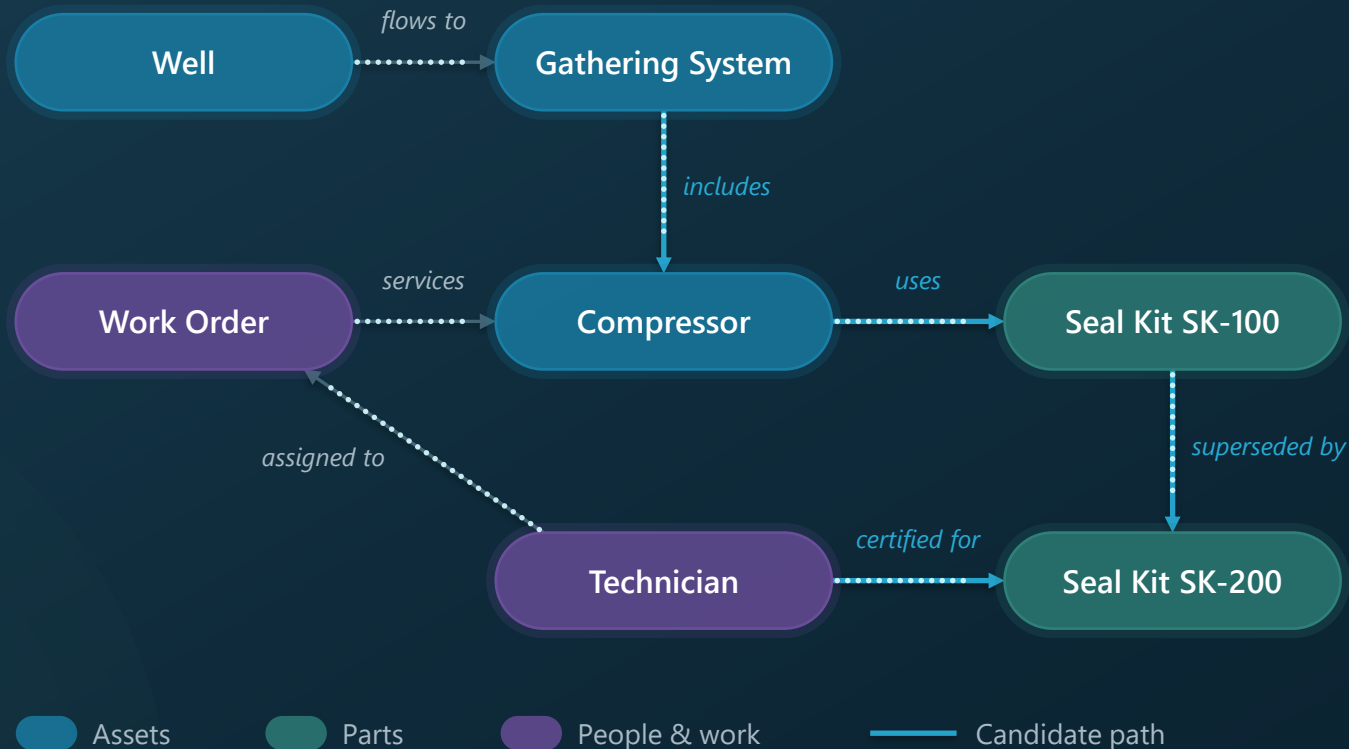
- **Separate memory types:** session, past interactions, ontology-typed asset facts, reusable skills
- **Track two timelines:** when true and when recorded. Supersede, never overwrite
- **Record provenance:** source, agent, reviewer
- **Right store for the job:** fast storage for sessions; a governed graph for facts
- **Ask “as of” questions** to reproduce audits and incident reviews

Prime the data before you retrieve it



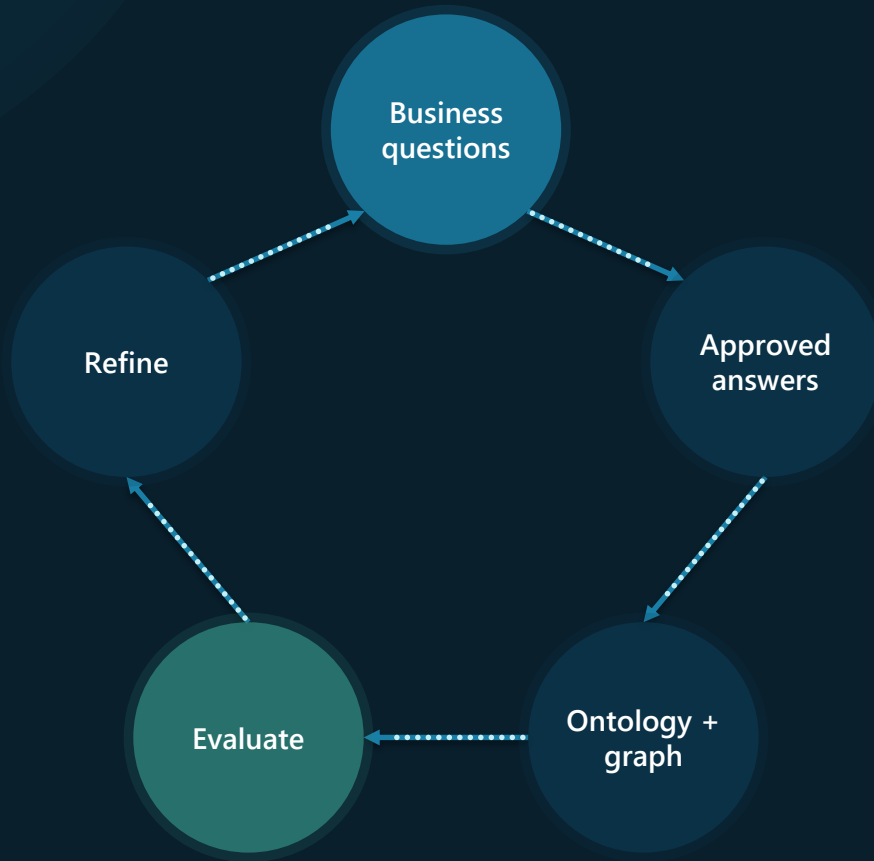
The ontology: one shared language for people and agents

Illustrative graph slice — candidate assets and personnel



- **Classes:** Field, Well, Equipment, Part, Work Order, Technician
- **Relationships:** flows to, uses, superseded by, certified for
- **Policy:** enforce approved substitution rules
- **Start from standards:** OSDU, PPDM, ISO 15926, then extend
- **Decision gates:** fit, service, dates, qualifications, authorization

Let business questions drive the ontology



Business-owned test set

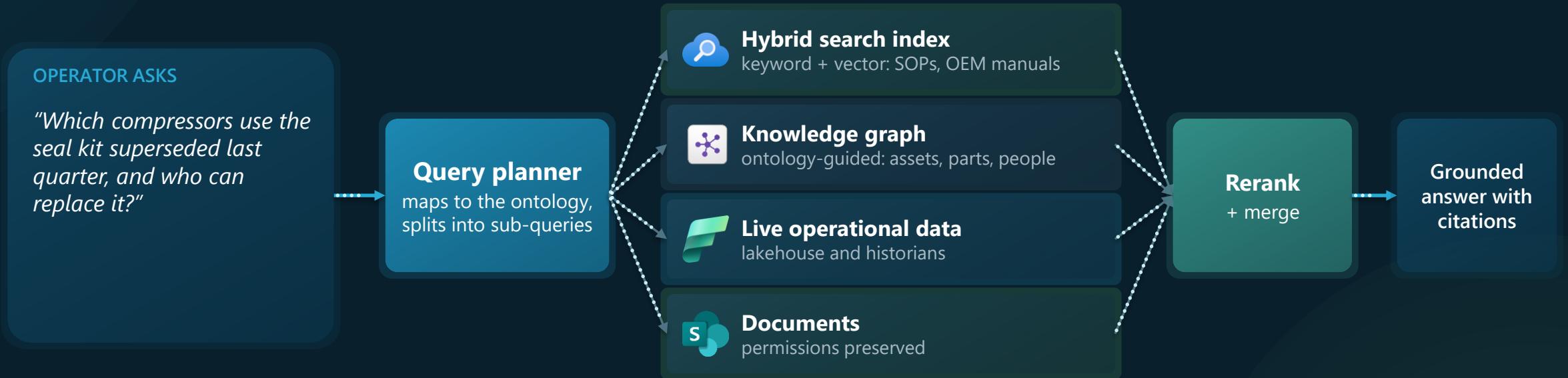
Production, maintenance and HSE leads own questions like "Which wells missed their integrity test?" and the approved answers.

Re-test as models improve

Newer models draft better ontologies. Re-run drafting and evaluation on every upgrade.

- **Log tokens and timings** per retrieval step
- **Show the evidence trail:** graph paths and documents behind each claim
- **Capture prompts and retrieved context** for debugging
- **Compare approaches side by side** on the same questions

One question, many silos: agentic hybrid retrieval



Break down multi-step questions

Run sub-questions in parallel across graph, search and live data.

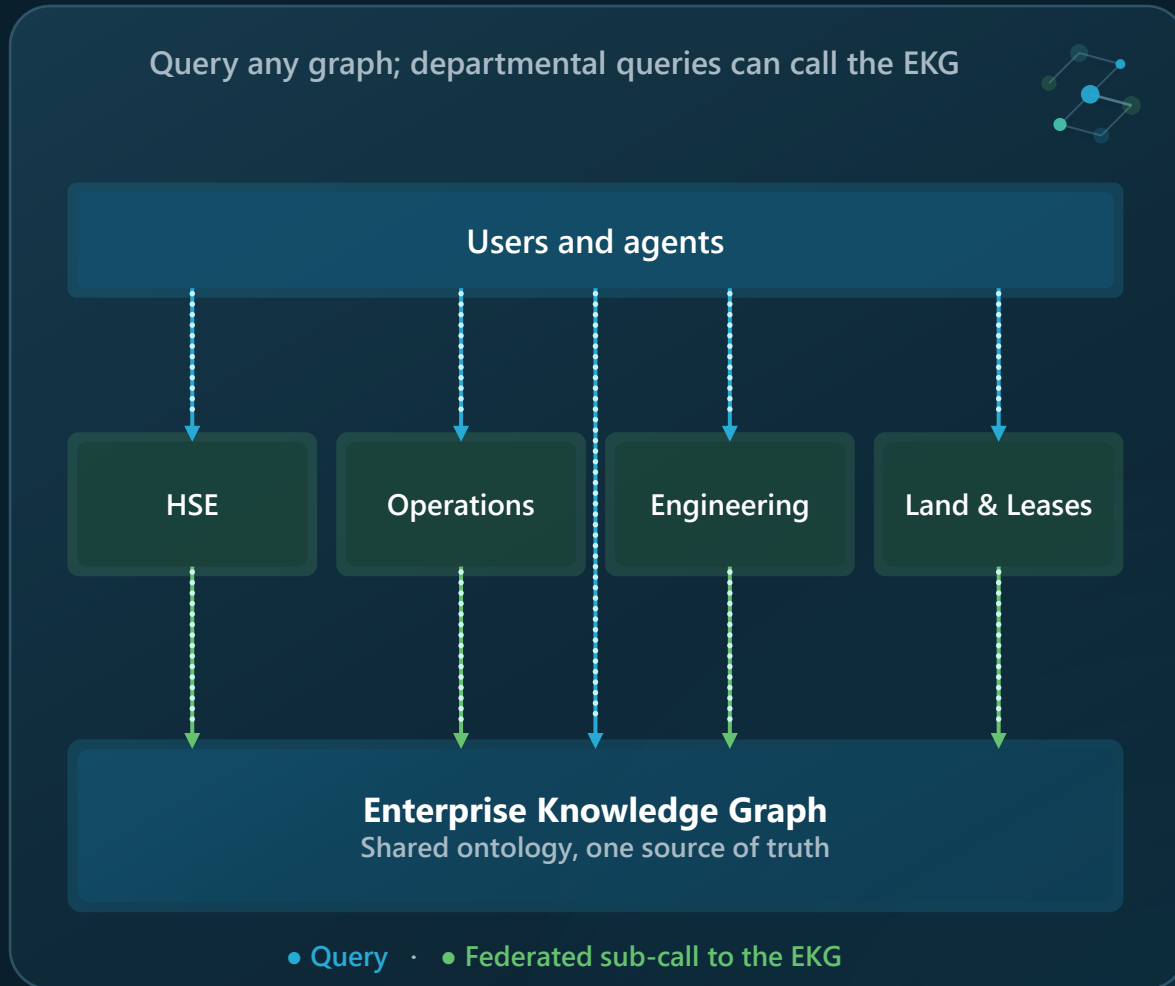
Tune effort per question

Dial reasoning effort; steer each source.

Fresh where it matters

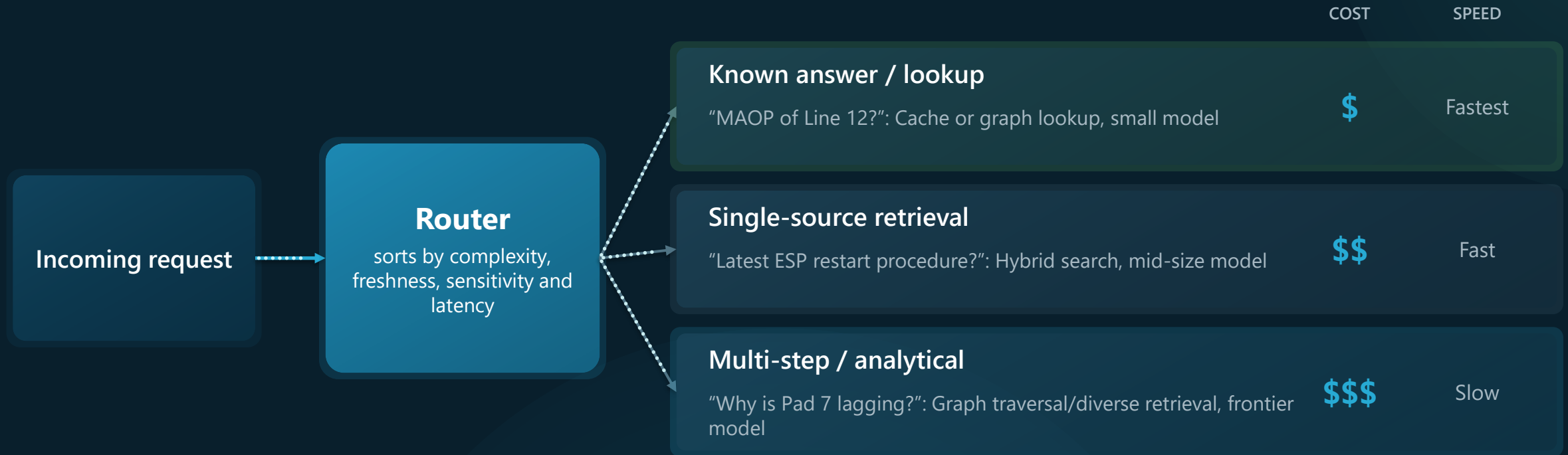
Query live data for status; index what changes slowly.

Least-privilege federated access to EKG



- **Enforce least privilege:** check user + asset scope at every retrieval hop
- **Virtualize data access:** some queries will only hit departmental graph, some will reach to the EKG
- **Carry identity end to end:** every agent, tool, index and federated call
- **Meet cyber-control expectations:** adhere to NERC CIP / TSA Security Directives where applicable
- **Keep source ACLs + audit trails:** private endpoints; log access and policy decisions.

Route the query, not just the model



Guardrails at the gateway

Caching, token limits and content safety in one place.

Escalate on low confidence

Start cheap; go deeper when evidence is weak.

Evaluate the router

Score routing on the same business questions.

Match the model and retrieval to the environment

Environment	Constraints	Routing pattern
Offshore and remote well sites	Spotty connectivity, limited compute	Small open-weight models on site; sync memory when connected
Field ops and control rooms	Low latency, high-consequence decisions	Advisory AI; authorized actions; trips and interlocks stay intact
Engineering and asset integrity	Multi-step questions, document-heavy	Graph traversal + frontier models; validate findings
Back office (land, JV accounting, supply chain)	High volume, sensitive data	Caching, mid-size models, permission-aware retrieval, burst GPUs

Reuse validated insights with governed write-back



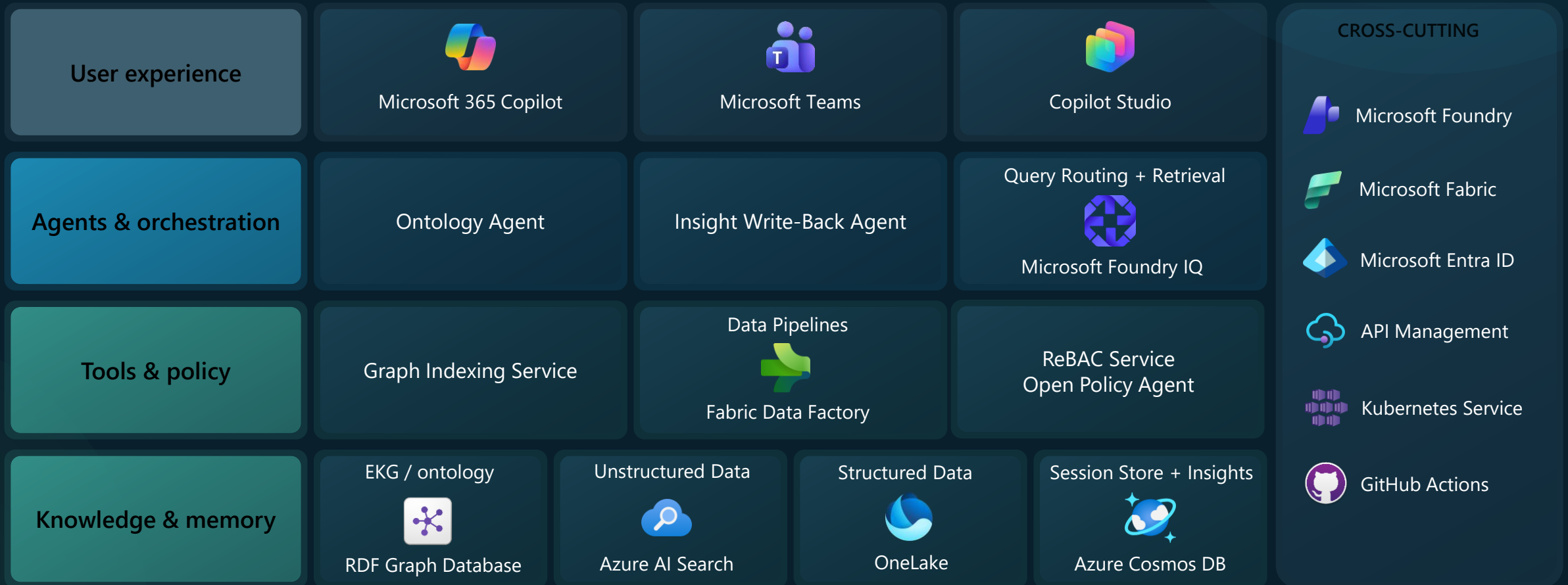
Why it matters

Reviewed observations, such as “Pad 7 seal failures track H₂S spikes,” retain evidence, scope and expiry. Reuse can reduce repeat analysis while valid; correlation is not causation.

Open challenges

- **Discoverability** of insights and skills
- **De-duplication** of overlapping insights
- **Evidence and trust** by source, scope and review status
- **Invalidation** of stale or unsupported insights

Reference architecture from the field*



* Enterprise knowledge graph deployment with an energy & utilities customer

Best-practice checklist

1 Context

- ✓ Treat context as a budget: plan, select, compress, cite
- ✓ Track when facts were true and when learned

2 Ontology & priming

- ✓ Let business questions drive the ontology
- ✓ Structure well files and P&IDs before indexing

3 Retrieval

- ✓ Decompose questions across graph, search and live data
- ✓ Enforce identity and permissions at retrieval

4 Routing

- ✓ Route by complexity, freshness, sensitivity and speed
- ✓ Measure cost, speed and evidence. Evaluate continuously

5 Your ontology is a business asset - fund and staff it like one.

- ✓ Talent gap: keep domain owners + an AI adoption lead in-house; partner for graph / AI engineering.

Thank you

Questions?

Context engineering · Ontologies & knowledge graphs

GraphRAG + agentic retrieval · Model and query routing

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This deck and other
Context Engineering
best practices